Table 15.1 Parameters in biological effects of radiation

Physical parameters
- Kind of radiation: α,β, or γ
- Internal or external source
- Size of total dose
- Dose rate
- Exposure time: instantaneous, temporary, recurrent, chronic

Microscopic effects
- molecular: DNA changes
- chromosome: abberations
- cellular (single or a few cells, *in vitro* or *in vivo*):
  - kind of cells (nervous system, bone marrow, liver, etc)
  - cell cycle (stage, rate of cell division, inactivation, etc)
  - temperature
  - oxygen content
  - cell poisons present (increasing radiation sensitivity)
  - protective agents present (anti-oxidants)
- tissue and organ: cancer growth or death

Macroscopic effects (inactivation, lethal dose (LD), cancer, etc.)
- Somatic: rapid (within a month), delayed (up to 20 years)
- Genetic: observed in offspring (one or several generations later)

15.1. THE BIOLOGICAL TARGET

The discussion of the effects of radiation on biological systems requires some familiarity with the target composition and common terminology.

The human body contains some 10^{14} cells. Figure 15.1.A shows schematically the cell and its nucleus, which contains thin thread-like *DNA-molecules* (*deoxyribonucleic acid*). The DNA carries the "genetic code" and is the most important molecule of an organism. It consists of two strands of sugar-phosphate chains, attached together by *base pairs* (forming so-called *nucleotides*), in a double helix form (Fig. 15.1.H–J). The human DNA contains 2.9×10^9 nucleotides; the DNA of simpler organisms has fewer nucleotides (down to about 5000). The nucleotides are combined in triplets (called *codons*), each one with the ability to produce (through some intermediate steps) a certain protein. The codons are ordered in long groups; the ordering is referred to as the *genetic code*. Because these long groups carry the information necessary to produce proteins for the different tissues, they are called *genes* ("makers"). In humans, the DNA is distributed over 23 pairs (altogether 46) *chromosomes*. In the chromosome, the DNA is wound around *histone* protein cores (Fig. 15.1.G) and highly twined ("condensed") to facilitate cell division; the DNA+histone unit (containing some 200 base-pairs) is referred to as the *chromatin*; these repeating units are known as *nucleosomes*. Figure 15.1.D shows some of the 23 chromosome pairs, and Figure 15.1.E a single one. The total amount of genetic

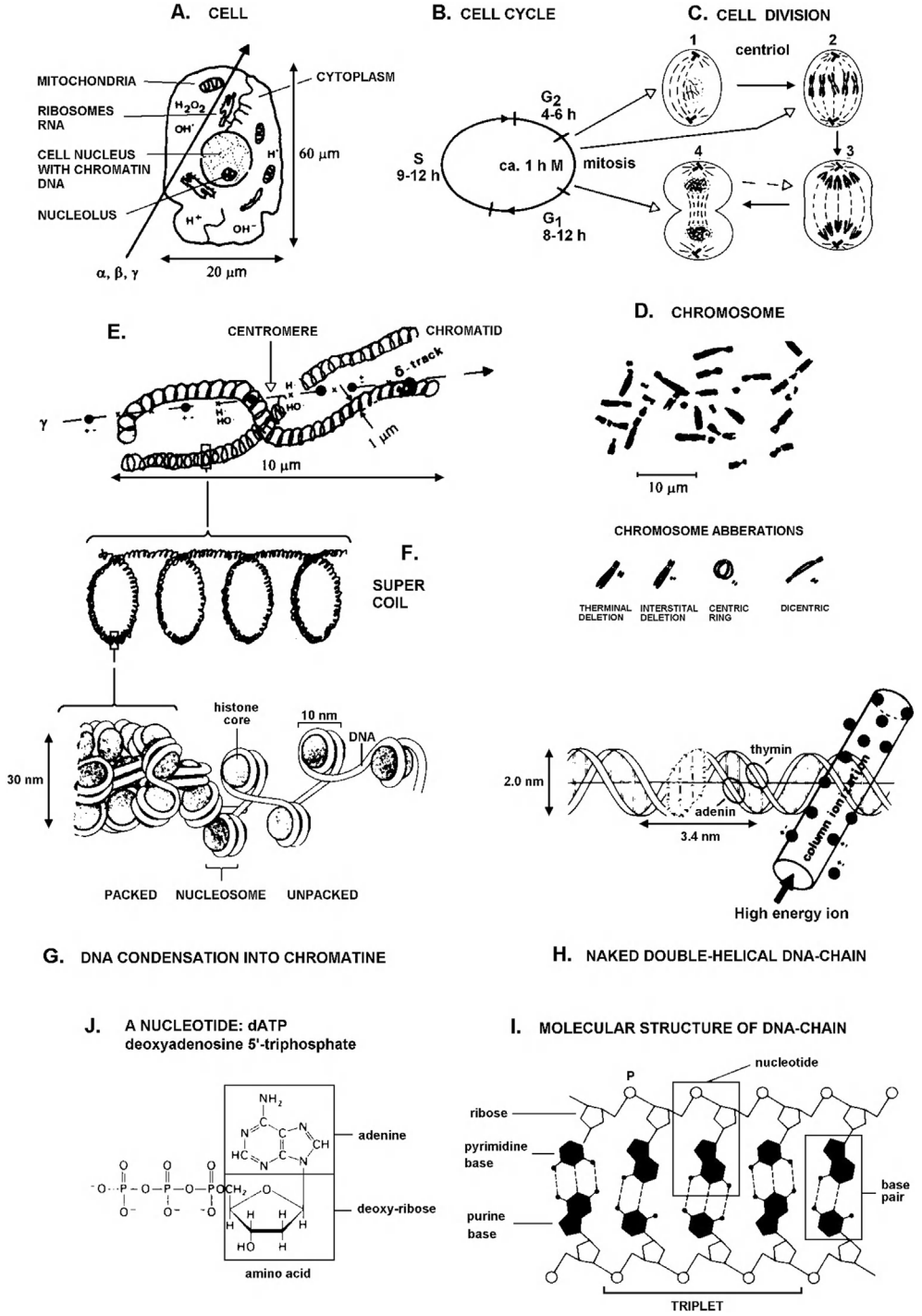


Figure 15.1 The cell and DNA biochemical system.

information in the cell is called the *genom*. Stretched out the total genetic material is about 1.5 m long, but its diameter is only 2 nm; the molecular weight is about 10^{11} per chromosome. While the cells comprise most of the body, the volume of the chromosomes only occupy ~1% of the cell volume and only about 1/10 of the chromosome volume carries a genetic message; thus the genetically significant target is rather small. Thus the probability that radiation should directly "hit" the DNA is small compared to the probability that radiolytically produced radicals will react chemically with the chromosomes.

The nucleotides are held together by hydrogen bonds between the nitrogen bases, which have their nitrogen rings perpendicular to the plane of Figure 15.1.I (i.e. turned 90° compared to the paper surface). The nitrogen rings contain some π-electrons, which interact between the planes and stabilize the chain structure. The unsaturated bonds in the nitrogen bases are sensitive to oxidation. Further, the phosphate oxygens are ionic. Thus, the DNA presents all types of chemical bonding and is sensitive to many types of reactants.

The *cell cycle* plays an important role in radiation damage (Figures 15.1.B and C) (i) during the S-stage proteins are synthesized, (ii) in the "gap 2", G_2, stage, the cell is being prepared for division (the DNA-chains are split up and copied), (iii) in the *cell division* (*mitosis*) the chromatin condenses into chromosomes, which are split and move in opposite directions, while the original cell is divided at mid-level (Figs. 15.1.C.3 and C.4), after which (iv) the cell "rests" or matures in the G_1-state.

15.2. RADIATION EFFECTS ON THE MOLECULAR LEVEL

When a high energy particle (whether α, β or γ) strikes a human cell it produces a narrow track, less than 1μm thick (cf. Figs. 7.5 and 8.1). The interaction with water, which is the main constituent of the cell, leads to radiolysis products, as described in Chapter 8. To a lesser extent, the radiation may interact directly with the molecules of the cell ("direct hit"). Only the effects (direct or indirect) on the cell nucleus DNA are thought to cause severe biological damage.

15.2.1. Radiation physics

High energy γ-rays lose most of their energy in a few compton scattering events. A "typical" natural radiation background gamma quantum of 1 MeV loses about 75% of its energy in 20 cm of water (a "typical" human thickness). This loss occurs to 50−80% in a single compton scattering event, producing an electron with up to 0.8 MeV kinetic energy, depending on the scattering angle. In the next compton interaction, the ≥0.2 MeV γ loses ≥50% of its energy; the distance between these initial two interactions exceeds cellular dimensions. Thus, the 1 MeV γ-absorption on the cellular level can be considered as the production of a single high energy electron. However, at very low